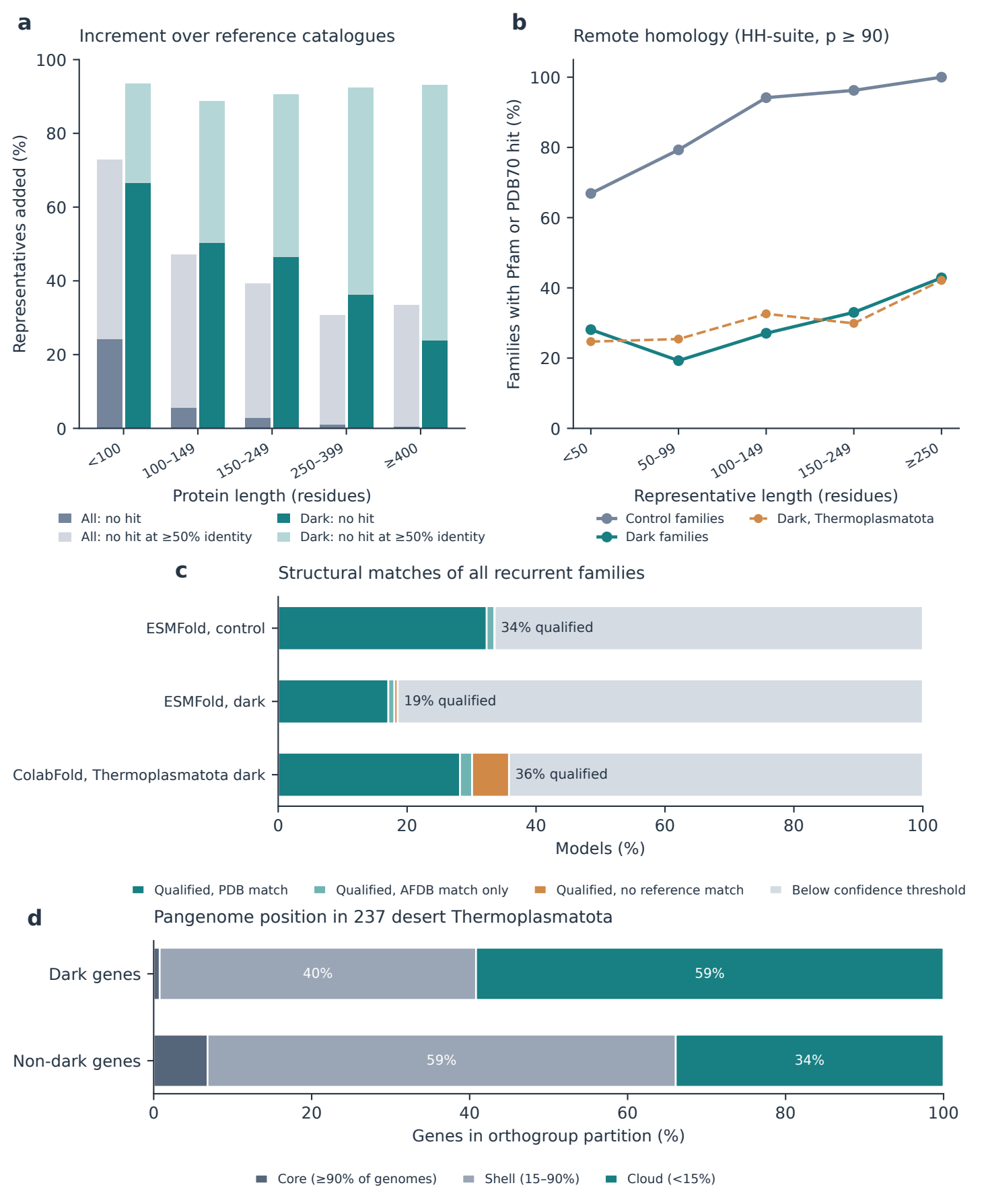
a
Increment over reference catalogues
Representatives added (%)
100
80
60
40
20
0
<100
100–149
150–249
250–399
≥400
Protein length (residues)
All: no hit
All: no hit at ≥50% identity
Dark: no hit
Dark: no hit at ≥50% identity
b
Remote homology (HH-suite, p ≥ 90)
Families with Pfam or PDB70 hit (%)
100
80
60
40
20
0
<50
50–99
100–149
150–249
≥250
Representative length (residues)
Control families
Dark families
Dark, Thermoplasmatota
c
Structural matches of all recurrent families
ESMFold, control
34% qualified
ESMFold, dark
19% qualified
ColabFold, Thermoplasmatota dark
36% qualified
0
20
40
60
80
100
Models (%)
Qualified, PDB match
Qualified, AFDB match only
Qualified, no reference match
Below confidence threshold
d
Pangenome position in 237 desert Thermoplasmatota
Dark genes
40%
59%
Non-dark genes
59%
34%
0
20
40
60
80
100
Genes in orthogroup partition (%)
Core (≥90% of genomes)
Shell (15–90%)
Cloud (<15%)